Rongfeng **Cheng** | Medical AI Research

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Research Profile

Undergraduate researcher at South China University of Technology studying whether medical AI models use clinically relevant evidence, how their behavior changes across datasets, and how physiological signals can be aligned with language. Two first-author manuscripts are under review at IEEE JBHI and IEEE BIBM 2026.

Education

South China University of Technology (SCUT) — **Guangzhou, China**
B.S. in Big Data Management and Application — *2023–2027*
GPA: 3.46/4.0.
Selected AI coursework: Large Models and Generative AI (93/100), Large Language Models and Prompt Engineering (93/100), Machine Learning (92/100), and Deep Learning (88/100).

Research Experience

Research Lead; First Author
Evidence-Use Auditing for Deep ECG Models — *Jan 2026–Present*
- Developed a matched-control perturbation framework to test whether deep ECG classifiers depend on clinically relevant evidence rather than task-correlated shortcuts.
- Evaluated PTB-XL and CODE-15% across architectures, perturbation doses, clinical evidence windows, and frozen-weight cross-dataset transfer.
- Manuscript under review at *IEEE Journal of Biomedical and Health Informatics* (JBHI).

Research Lead; First Author
DLEF: Dataset-Label Evidence Fingerprints — *Jun 2026–Present*
- Proposed directional and boundary-aware evidence fingerprints to characterize dataset-label evidence structure.
- Studied cross-dataset AUROC degradation across CODE-15%, PTB-XL, and Chapman-Shaoxing.
- Manuscript under review at IEEE BIBM 2026.

Research Lead
PathCLIP: Knowledge-Aware ECG–Text Representation Learning — *Jul 2026–Present*
- Investigating diagnostic hierarchies, ECG findings, and evidence grounding for ECG–text alignment and zero-shot recognition.
- Designed matched-count, permuted-hierarchy, residual-only, and held-out-label controls; current work shifts from ontology-only supervision toward finding-level evidence alignment.

Publications

First author: **Auditing Knowledge-Aligned Evidence Use in Deep Learning Models: A Matched-Control Study in ECG Classification.** Under review at IEEE JBHI.

First author: **DLEF: Dataset-Label Evidence Fingerprints for Cross-Dataset ECG Generalization.** Under review at IEEE BIBM 2026.

Co-author: **CodeSafetyBench: Evaluating and Improving Ethical Safety in Large Language Model Code Generation.** *iScience*, 2026. doi:10.1016/j.isci.2026.115073.

Co-author: **Artificial Intelligence Applications in X-ray Coronary Angiography: A Comprehensive Review of Image Analysis Techniques.** HNNDL 2026. doi:10.1145/3795892.3795917.

Co-author: **Artificial Intelligence for Echocardiographic Assessment of Pulmonary Hypertension and Right Heart Function: Advances in Intelligent Diagnosis and Risk Stratification.** HNNDL 2026. doi:10.1145/3795892.3795918.

Professional Experience

AI Research Institute, Guangdong No. 2 Provincial People's Hospital — **Guangzhou, China**
Research Intern — *Nov 2024–Present*

- Conduct research on medical LLMs, cardiovascular knowledge representation, clinical question answering, and AI product prototyping.
- Contributed to a standardized-patient Q&A pipeline built from about 8,000 clinical cases, a myocardial-ischemia knowledge graph, patents, software copyrights, and clinical AI prototypes.

Selected Projects

Medical LLM / Knowledge Representation
KG-Based Standardized Patient Q&A
Processed approximately 8,000 cardiovascular case PDFs and built an LLM-based extraction pipeline with named-entity recognition, relation extraction, entity disambiguation, Neo4j storage, and hybrid RAG.

Full-Stack Developer; Assessment-Scale Co-Designer
Dingbei Meimei – Acne & Facial Health AI Assessment System
Built the complete front end and back end, including iPad adaptation and the clinical case workflow; co-designed the assessment scales with dermatologists at Guangdong No. 2 Provincial People's Hospital. Delivered the system for the July 2026 launch of the hospital's Refractory Acne Diagnosis and Treatment Center.

Project Lead
Myocardial Ischemia Knowledge Graph
Built a domain knowledge graph with collaborative ontology management, XGBoost-based implicit reasoning, and graph-mapped interpretability analysis.

Clinical AI / MIMIC-IV
IHD Trajectory Similarity Retrieval
Developed trajectory retrieval and post-cutoff validation for ischemic heart disease ICU stays using time-windowed clinical variables and precomputed k-NN similarity graphs.

Patents & Intellectual Property

- Method, System, and Electronic Device for Medical Question-Answering Output Assurance – First Inventor; application #202610582199.6; filed Apr 2026; preliminary examination passed.
- Knowledge Graph Ontology Intelligent Management System V1.0 – Software Copyright #2025SR2454864.
- LLM-Based Cardiovascular Proactive Health Management Intelligent System V1.0 – Software Copyright #2025SR2080673.

Honors

- First Prize, 2025 Challenge Cup (Jinan University Campus).
- Second Prize, 2025 9th Guangzhou Mathematical Modeling University League.
- Third Prize, 2025 National Mathematical Modeling Contest – Guangdong Division.
- National Innovation and Entrepreneurship Training Program – Project Completed.

Skills

Programming: Python, TypeScript, SQL
AI / ML: PyTorch, representation learning, multimodal learning, XGBoost
Medical AI: ECG analysis, evidence-use auditing, cross-dataset evaluation, clinical LLMs
Web / Data: Next.js, Flask, Neo4j, Git, Docker

Languages

Chinese (native)   Cantonese (native)   English: IELTS 6.5, CET-6 528